

HPS Gaussian-Process Resonance Search

Reference Matching and Fixed-Yield Injections
A short statistical study, v6.3

Emrys Peets

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1 Answer and the two questions being tested

Per-toy reference matching is statistically legitimate as a declared, adaptive injection–recovery diagnostic. It does not, by itself, test coverage or detection power at one fixed physical signal yield. For that purpose, use **one common expected yield per mass and dataset**, preferably chosen from an independent background-only pilot sample and then frozen. The existing matched study remains useful for understanding response and signal absorption; changing the design does not automatically cure bias.

Let B_t be background-only pseudoexperiment t , and define

$$s_t = \sigma_{A,\text{ref},t} = s(B_t), \quad u_t = \hat{A}_t(0), \quad e_t(A) = \hat{\sigma}_{A,t}(A). \quad (1)$$

Here s_t is the fitted uncertainty on the *signal yield* before adding signal, u_t is the signed zero-signal fitted yield, and $e_t(A)$ is the uncertainty returned after injection. These are not a mass resolution or an uncertainty on the ensemble mean. For $z \in \{1, 3, 5\}$, compare

$$\boxed{A_t^{\text{M}} = z s_t} \quad (\text{per-toy matched}), \quad \boxed{A^{\text{F}} = z s_0} \quad (\text{common expected yield}). \quad (2)$$

In the proposed alternative, s_0 estimates the mean reference uncertainty within a specified cell. A denotes the *expected* signal count; a Poisson realization need not contain exactly A signal candidates.

In plain words. Matching gives a noisier toy a larger injected signal. A common yield gives every toy the same expected number of signal candidates, allowing the difficulty of extracting it to fluctuate naturally.

Input z	Matched: $s_t = 10$	Matched: $s_t = 20$	Common: $s_0 = 15$
1	10	20	15
3	30	60	45
5	50	100	75

Table 1: Illustrative yields in candidates, with the two reference errors equally likely. Both designs have the same average yield, but different experiments.

“Matched” also means using the same binning, template, mask, fit procedure and error convention for the reference and extraction. Keep that procedural consistency in *both* designs. It does not require A to vary by toy.

2 Why equal average yield does not make equal ensembles

For this comparison, first take the ideal common scale $s_0 = \mathbb{E}[s_t]$. The mean injected yield agrees, but its variation does not:

$$\begin{aligned} \mathbb{E}[A_t^M] &= A^F = z s_0, & \text{Var}(A_t^M) &= z^2 \text{Var}(s_t), & \text{Var}(A^F) &= 0, & (3) \\ \frac{\text{SD}(A_t^M)}{\mathbb{E}[A_t^M]} &= \text{CV}(s_t), & \frac{A_t^M}{s_t} &= z, & \frac{A^F}{s_t} &= z \frac{s_0}{s_t}. & (4) \end{aligned}$$

The last two ratios are nominal input strengths. Neither is automatically the fitted significance, a discovery significance or a global significance. An injected signal can alter the error and the fitted background.

Let P_0 generate the background and K_A generate added signal counts at expected yield A . The actual sampling laws are

$$P_M(dB, dS) = P_0(dB) K_{zs(B)}(dS), \quad (5)$$

$$P_F(dB, dS) = P_0(dB) K_{zs_0}(dS). \quad (6)$$

In the matched design, the signal expectation is a function of the same background fluctuation that will be fitted. It is generally *not* an ordinary mixture of fixed- A experiments: selecting a particular $s(B)$ also selects background realizations. If s_t were constant, the two laws would coincide. A small spread makes their input yields close, but does not prove that nonlinear fit response or tail probabilities are equivalent.

Coverage and power have a fixed-parameter meaning

For a confidence interval $I(Y)$ and a rejection rule $\phi(Y)$, the usual quantities at fixed background nuisance parameters ν are

$$C(A, \nu) = \Pr_{A, \nu}\{A \in I(Y)\}, \quad \pi(A, \nu) = \Pr_{A, \nu}\{\phi(Y) = 1\}. \quad (7)$$

The standard frequentist construction repeats the experiment at fixed parameter values, generating the relevant auxiliary measurements as well as the main data when they are part of the procedure [1]. Matched-toy containment instead estimates

$$C_M = \Pr_M\{zs(B) \in I(B + S)\}. \quad (8)$$

This is a well-defined performance measure for the adaptive experiment. It is not a measurement of $C(zs_0, \nu)$.

In plain words. For a coverage claim, the truth must stay put while the experiment fluctuates. If the truth moves with each toy, a containment fraction answers a different question.

This distinction does not imply that matched pulls *must* fail. In an ideal location model $\hat{A} = A + s_t E_t$ with $E_t \sim N(0, 1)$ independent of s_t and a correctly returned error s_t , the pull is E_t for either design. Matching neither forces a biased estimator to close nor proves fixed-yield performance when closure is observed.

3 Separate background offset, signal response and error calibration

To expose the mechanism, write the local conditional-mean response as

$$\mathbb{E}[\hat{A}_t(A) \mid B_t] = u_t + g_t A + O(A^2). \quad (9)$$

The signal draw is averaged on the left. u_t is the background-dependent offset; $g_t = 1$ means unit incremental response and $g_t < 1$ means a loss of injected yield in this approximation. Ignoring the displayed nonlinear term, the ensemble mean biases are

$$b_M = \mathbb{E}[u_t] + z\mathbb{E}[(g_t - 1)s_t], \quad (10)$$

$$b_F = \mathbb{E}[u_t] + zs_0(\mathbb{E}[g_t] - 1), \quad (11)$$

$$\boxed{b_M - b_F = z \text{Cov}(g_t, s_t)} \quad (s_0 = \mathbb{E}[s_t]). \quad (12)$$

Thus a correlation between sensitivity and absorption changes the average bias even when the mean injected yield agrees. The coefficient of variation of s_t alone cannot determine that correlation.

Three quantities that should be reported together

For $A_t > 0$, define the raw recovery, paired response and pull as

$$Q_t = \frac{\hat{A}_t(A_t)}{A_t}, \quad R_t = \frac{\hat{A}_t(A_t) - \hat{A}_t(0)}{A_t}, \quad p_t = \frac{\hat{A}_t(A_t) - A_t}{e_t(A_t)}. \quad (13)$$

Raw recovery includes the background offset. The paired response subtracts that offset to isolate the response to adding signal. In the linear model, $\mathbb{E}[R_t \mid B_t] = g_t$ in either design; an individual Poisson signal draw makes R_t noisy. A raw average of Q_t additionally contains $\mathbb{E}[u_t/s_t]/z$ under matching, or $\mathbb{E}[u_t]/(zs_0)$ at common yield. Even $\mathbb{E}[u_t] = 0$ does not imply $\mathbb{E}[u_t/s_t] = 0$.

Do not interchange the average of per-toy ratios and the ratio of averages. For matched injections in this linear model,

$$\mathbb{E}[R_t] = \mathbb{E}[g_t], \quad \frac{\mathbb{E}[\hat{A}_t(A_t) - \hat{A}_t(0)]}{\mathbb{E}[A_t]} = \frac{\mathbb{E}[g_t s_t]}{\mathbb{E}[s_t]} = \mathbb{E}[g_t] + \frac{\text{Cov}(g_t, s_t)}{\mathbb{E}[s_t]}. \quad (14)$$

Design	Mean A	Mean bias	Mean Q_t	Mean R_t
Per-toy matched	45	−6.30	0.923	0.890
Common yield	45	−4.95	0.890	0.890

Table 2: Exact illustrative example at $z = 3$: equally likely states have $s = (10, 20)$, $u = (4, -4)$ and $g = (0.98, 0.80)$. Bias is in candidates. The difference is $3 \text{Cov}(g, s) = -1.35$. These are not HPS measurements.

Always use each injected fit’s returned error in the pull; replacing it by the pilot mean would test a different error prescription. Pull moments are diagnostics, not substitutes for interval coverage. For physical inference, retain the raw fitted yield; paired subtraction is a response diagnostic.

At $z = 0$ both designs inject zero. With the same background draws and fitting procedure, their null distributions are identical. Changing injection scaling cannot remove a null offset. Also distinguish a signed local estimator from a positive, one-sided scan maximum; v5.9.5 addresses these separately [4].

4 What the archived HPS implementation actually did

The v5.0.5 methodology defines a zero-signal reference fit for each source, toy and mass, followed by $A_t = z\sigma_{A,\text{ref},t}$ and a refit of the injected spectrum [3]. The audit for this report follows the preserved v4.6 full-100 baseline used in the note’s consolidated validation displays. It covers four 2021 exposure/source constructions and five masses (65, 90, 120, 180, 210 MeV), with 7,994 accepted extraction rows. It does not pool the later targeted threshold replacements.

The saved rows and generating code establish the following:

- The actual injection scale is the selected background-only fit’s returned `sigma_A`, saved as `sigmaA_ref`. The reference fit uses that toy’s fluctuated counts, including its held-out extraction region. It is not a universal Asimov scale.
- All 5,995 accepted positive-injection rows satisfy $A_t = z s_t$, to an absolute numerical residual below 3×10^{-11} candidates. Their saved reference error equals the corresponding zero-injection returned error exactly.
- Added signal counts are **Poisson fluctuated**. Across the accepted rows, $(N_{\text{sig}} - A_t)/\sqrt{A_t}$ has descriptive sample standard deviation 1.005. The code, rather than this pooled check alone, establishes the sampling mode.

Mass [MeV]	Mean s_t	SD(s_t)	CV [%]	Corr(s_t, u_t)
65	7,327.5	206.2	2.814	+0.162
90	6,494.3	154.4	2.377	-0.344
120	5,055.3	90.7	1.794	-0.065
180	2,853.6	53.9	1.890	+0.079
210	2,144.0	43.5	2.027	+0.180

Table 3: Historical 2021 10% baseline, 100 reference toys per mass. Errors and their standard deviations are in signal candidates. CV is $\text{SD}(s_t)/\bar{s}$. The last column is the descriptive sample correlation with the signed zero-injection yield. It is not a significance test.

Across all 20 historical cells, the CV ranges from 1.17% to 3.00%. Consequently, the matched *expected* yields vary by that relative amount; this excludes the additional Poisson variation of actual signal counts. At 65 MeV in the 2021 10% row, a common scale equal to the saved mean would give approximately 7,327, 21,982 and 36,637 expected candidates at $z = 1, 3, 5$. These are illustrative translations of an existing ensemble, not new fixed-yield fit results.

In plain words. In this archived baseline the two injection choices are numerically quite close in yield. The audit does not show that reference matching caused the observed bias, and it cannot reconstruct how new fixed-yield fits would behave.

There are two naming traps. On zero-injection rows, `sigmaA_reference` is a *separate Asimov diagnostic*; using that column would audit the wrong quantity. Also, the older v4.6 plotting code labels $\hat{A}/A_{\text{inj}}$ as recovery, whereas the v5.0.5 methodology defines the paired response in Eq. (13). The audit preserves and computes both definitions without relabeling historical plots.

5 How to choose the common reference uncertainty

For a fixed dataset, mass, template and fit prescription, use an independent pilot ensemble of M background-only fits:

$$s_0 = \bar{s}_{\text{pilot}} = \frac{1}{M} \sum_{j=1}^M s_j^{\text{pilot}}, \quad A \in \{0, s_0, 3s_0, 5s_0\}. \quad (15)$$

Freeze that value before generating the evaluation toys. Conditional on the frozen pilot result, every evaluation toy then has the same expected signal yield. Pilot precision is approximately

$$\frac{\text{SE}(\bar{s}_{\text{pilot}})}{\mathbb{E}[s]} \simeq \frac{\text{CV}(s)}{\sqrt{M}}. \quad (16)$$

This quantifies precision of the chosen injection coordinate; it does not include uncertainty in the physical background truth.

A mean over the *same* evaluation backgrounds is usable as an exploratory common-yield check, but retains a weak dependence on those backgrounds. For independent, identically distributed background toys and a baseline statistic $u_t = u(B_t)$,

$$\text{Cov}(z\bar{s}_T, u_t) = \frac{z}{T} \text{Cov}(s_t, u_t). \quad (17)$$

It shrinks with ensemble size, unlike the per-toy dependence $\text{Cov}(zs_t, u_t) = z \text{Cov}(s_t, u_t)$. The independent pilot removes this particular dependence; it is the cleaner design for a fixed-yield test.

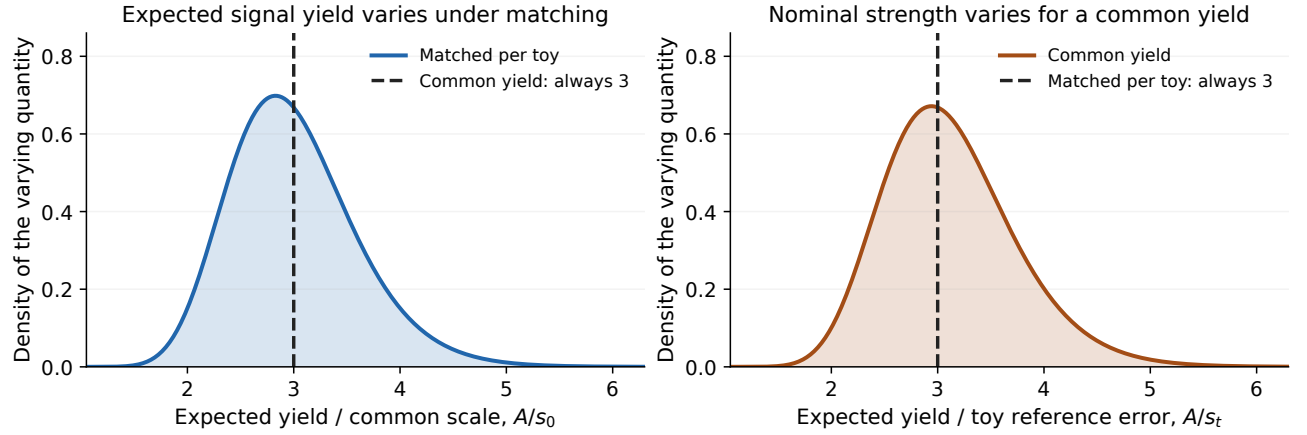


Figure 1: Analytic illustration at $z = 3$, with a positive lognormal reference error having mean s_0 and CV 20%. The broad spread is chosen for visibility and is *not* the measured HPS spread. Curves are densities of the varying quantity; dashed lines mark a constant value, not a density height. Matching fixes A/s_t and varies A ; a common yield fixes A and varies A/s_t .

An Asimov background-only fit also supplies a common scale; its curvature error need not equal the mean error of fluctuated fits [2]. For a target median local significance, tune a fixed A with the Asimov approximation and verify it with toys. A/s_0 alone need not equal that significance.

The mean error differs from both $\sqrt{\mathbb{E}[s_t^2]}$ and $\text{SD}(\hat{A}_t(0))$. The last measures estimator variation; averaging errors does not establish their calibration. Do not pool masses, datasets or incompatible yield normalizations. To compare extraction methods, use one nominal scale and the *same injected spectra*.

6 A separate choice: fluctuate the number of signal candidates

Choosing a common A does *not* require adding exactly A candidates to every toy. Let w_i be full-template signal probabilities and include out-of-support categories so that $\sum_i w_i = 1$. Let $B_i \sim \text{Poisson}(b_i)$ independently. Three useful designs have different sampling variances:

Signal construction	Mean and covariance	Question it addresses
Deterministic template	$S_i = Aw_i$, $\text{Cov}(S) = 0$	Mean-response and absorption control; omits signal counting noise.
Exactly N candidates	$S \sim \text{Multinomial}(N, \mathbf{w})$; $\mathbb{E}[S] = N\mathbf{w}$, $\text{Cov}(S) = N[\text{diag}(\mathbf{w}) - \mathbf{w}\mathbf{w}^T]$	Recovery of a specified injected sample size; the total count is fixed.
Poisson signal yield	$S_i \sim \text{Poisson}(Aw_i)$ independently; $\text{Cov}(S) = A \text{diag}(\mathbf{w})$	Repeated counting experiments at a fixed expected signal rate A .

Table 4: The background counting covariance adds to these covariances when the fixed-yield signal draw is independent of the background. For matched injections these signal formulas apply conditional on B ; marginally the adaptive dependence must also be retained.

Drawing $N_{\text{sig}} \sim \text{Poisson}(A)$ and then placing those candidates multinomially is equivalent to independent binwise Poisson draws. In the fixed-yield rate experiment,

$$B_i + S_i \sim \text{Poisson}(b_i + Aw_i). \quad (18)$$

By contrast, an exactly- N signal has negative cross-bin covariances and no fluctuation in its total. Deterministic addition has no signal counting variance at all. A Poisson extraction likelihood applied to these other ensembles need not produce unit pull widths or nominal interval coverage, even with the correct background mean.

In plain words. “Inject the same signal” should usually mean the same expected rate. Real counting experiments fluctuate around that rate. Adding exactly the same number of candidates is a useful controlled experiment, but it removes one source of fluctuation.

For Poisson signal under per-toy matching, total signal variance satisfies

$$\text{Var}(N_{\text{sig}}) = \mathbb{E}[A_t] + \text{Var}(A_t) = z\mathbb{E}[s_t] + z^2 \text{Var}(s_t). \quad (19)$$

At a common expected yield it is just A . The extra term is variation of the chosen truth between toys; it is not an additional detector uncertainty. Pulls must subtract each toy’s own expected A_t , not an ensemble average or that toy’s realized N_{sig} , when assessing a Poisson-rate estimator.

Relation to v6.2

The current v6.2 MC study already uses common sample sizes $N = 1,000, 5,000, 10,000, 30,000$ at each mass, with exactly- N multinomial signal and Poisson background [5]. It is a fixed-count recovery study, rather than a new per-toy reference-matched scan. Its GP conditional mean and covariance are recomputed, while archived kernel hyperparameters remain fixed. For a Poisson-rate coverage follow-up, keep a common *expected* A and also fluctuate the total signal count. In either case, repeat whichever fit procedure the claimed result actually uses.

7 Recommended next test and what this report establishes

1. **Define the cell and yield.** Fix dataset, mass, full signal template, generating background and extraction prescription. Preserve the production $\pm 2.25\sigma_m$ exclusion window. State whether A is a full selected yield or a yield restricted to a specified region.
2. **Freeze a common scale.** Use an independent background-only pilot mean, or a declared Asimov scale, then take $A = 0, s_0, 3s_0, 5s_0$. Use the same absolute yields when comparing methods or testing robustness to alternative generating backgrounds; give any separate equal-difficulty comparison its own label.
3. **Generate evaluation toys.** For rate inference draw independent Poisson background and signal counts. Sharing each background across injection levels provides useful paired controls. Fit identical spectra with competing methods; account for the pairing when estimating differences.
4. **Keep the three diagnostics.** Report signed raw bias, per-fit-error pulls, and paired incremental response. Report failed fits and their selection rule. Do not discard statistical outliers or tune a bias correction on the evaluation sample.
5. **Measure coverage or power separately.** At each fixed A , evaluate the actual interval construction or test, with binomial uncertainty on the resulting fraction. Repeating a fitted, frozen background source gives a source-conditional result; it does not establish uniform performance over unknown physical backgrounds. Keep the matched design as a secondary response check.

Decision. The proposed average-error injection is a sensible primary design for a common-yield validation, with an independent pilot as the clean implementation. There is no basis here to discard the older matched study as statistically invalid. It asks a different question, and the archived baseline has only a few percent variation in its matched expected yields. Neither design alone validates the physical background or fixes null bias.

Scope, audit trail and reproduction

This study derives the comparison and audits archived rows; it runs **no new HPS fits** and estimates no new coverage or discovery probability. The package includes LaTeX, the analytic figure, minimal archived rows, scripts, CSVs and SHA-256 provenance. Run `bash reproduce.sh`; see `README.md` for tool paths. The original row source has SHA-256 prefix `b23f31975b50`; `audit/` contains full hashes and line evidence. Later threshold replacements are outside this quantitative audit.

References

- [1] G. Cowan and L. Lista, “Statistics,” *Review of Particle Physics* (2026), especially Secs. 40.2.6.1 and 40.4.2. PDG Statistics review.
- [2] G. Cowan, K. Cranmer, E. Gross and O. Vitells, “Asymptotic formulae for likelihood-based tests of new physics,” *Eur. Phys. J. C* **71**, 1554 (2011), Secs. 3.2 and 4.1. arXiv:1007.1727.
- [3] HPS-GPR analysis note v5.0.5 (16 September 2026), “Matched-reference injection–extraction diagnostics” and “Matched refits and fit-quality checks.” Preserved v4.6 full-100 baseline code and accepted rows are identified in the accompanying audit.
- [4] HPS-GPR v5.9.5, *Apparent null bias* (22 September 2026), local study package.
- [5] HPS-GPR v6.2, *Reconstructed-MC injection and signal recovery* (23 September 2026), local study package, 40-toy release.